

MYSQL DATABASE MANAGEMENT SYSTEM

LAB SHEET #8

Title:

- Implement further sub queries with DML commands.

Objectives:

- To understand and implement concept of sub queries.
- To understand and use nested sub query in DML Commands like SELECT, INSERT, UPDATE, DELETE.

Background Concept:

DB Source: https://github.com/datacharmer/test_db

- **CREATE TABLE FROM EXISTING TABLE**
 - CREATE TABLE <new_table_name> LIKE <existing_table>;
 - CREATE TABLE <new_table_name> AS SELECT <column_list> FROM <existing_table> WHERE 1 = 0 ; (for copying structure only without records)
- **SUB QUERY**
 - **INSERT**
 - INSERT INTO <table_name> (<column_list>) SELECT <column_list> FROM <existing_table> WHERE <condition>;
 - **UPDATE**
 - UPDATE <table_name> SET <column_name> = <value> WHERE <column> <operator> (SELECT <column_list> FROM <existing_table> WHERE <condition>);
 - **DELETE**
 - DELETE FROM <table_name> WHERE <column> <operator> (SELECT <column_list> FROM <existing_table> WHERE <condition>);

Lab works:

1. Find employee no., name and salary of those employees earning more than average salary.
2. Find employees working in 'Sales' department using sub queries.
3. Find employees who are department managers.
4. Create a new empty table named 'high_salary_emp' just like employees.
5. Insert record of employees earning more than 80,000.
6. Create a new empty table 'sales_emp' which is similar as employees table.
7. Insert records of employees from sales department only.
8. Create a duplicate table of 'salaries' table as 'new_salaries' along with data.
9. Increase salary by 10% for employees in Development department.
10. Increase salary of employees by 5000 whose earning is below average salary.
11. Delete salary records which are below average salary.
12. Create duplicate table for 'employees' table as 'new_employee'.
13. Delete records of those employees who are not assigned to any departments.
14. Delete all the newly made tables.